

# Mathematics A

## Unit 1 • Chapter OL-T33 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 6 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

**Dr Eslam Ahmed**

Elite IGCSE Mathematics • [eliteigcse.com](https://eliteigcse.com)

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

# Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T33 • Expertise Q16+ • 2 items • 6 marks

| Chapter                                        | Items | Marks | Starts |
|------------------------------------------------|-------|-------|--------|
| 01. OL-T33 <a href="#">Graphs of Functions</a> | 2     | 6     | 4      |

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

# Graphs of Functions

Unit 1 • Expertise • 6 marks • 2 item(s)

June 2025 | 4MA1-2H Q25 | 3 marks

Feature-led polynomial sketch

June 2023 | 4MA1-2H Q18 | 3 marks

Recognise reciprocal or rational graph shape

Original question context and synchronized companion pages follow.

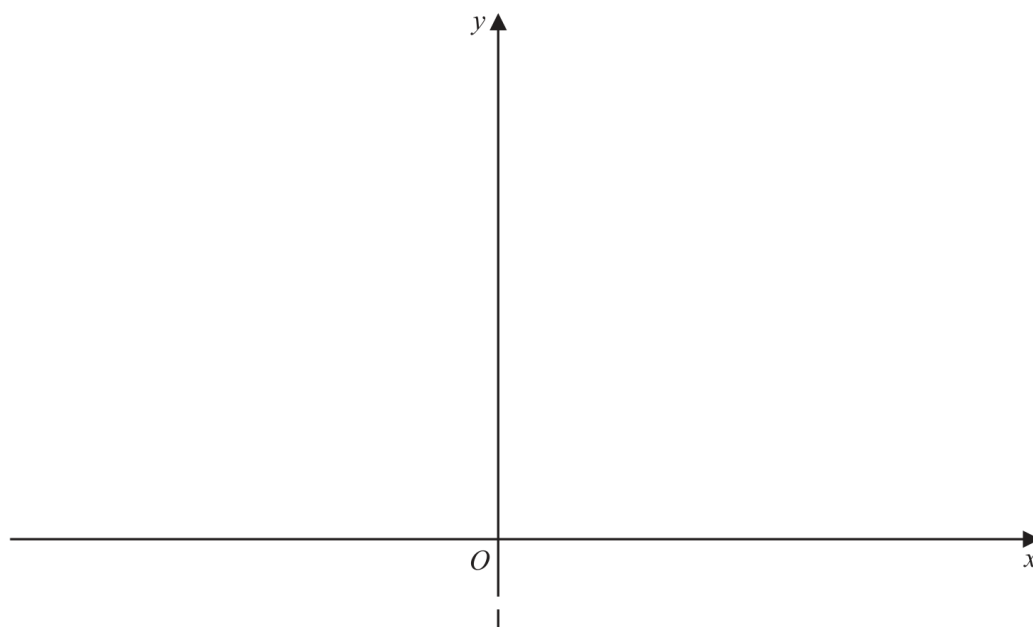
Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q25

3 marks

(b) On the axes below, sketch the graph of  $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the  $y$ -axis.



(3)

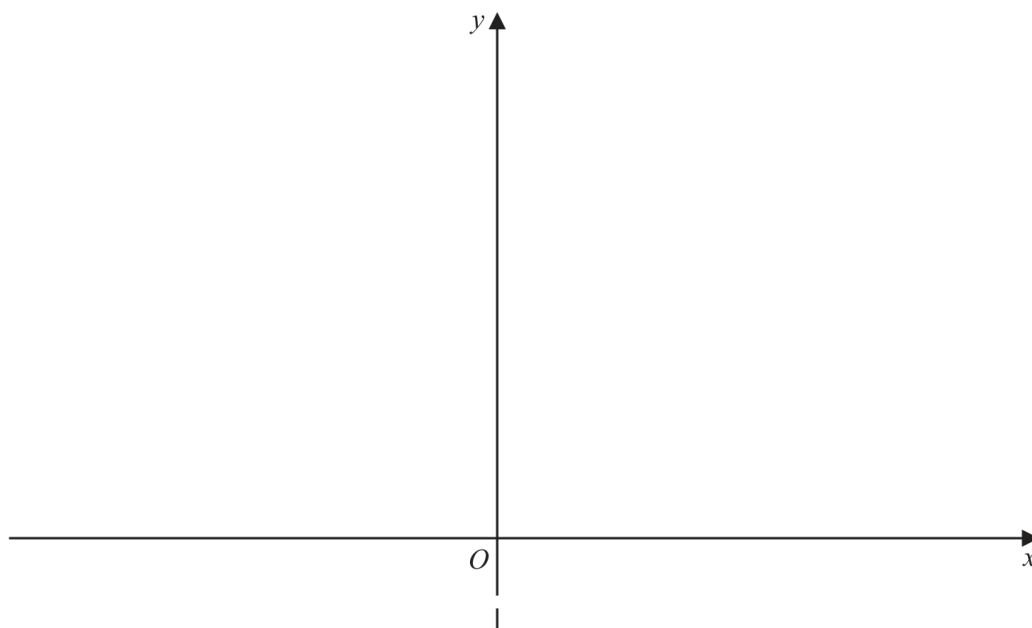
WORKING SPACE

4MA1-2H Q25

3 marks

(b) On the axes below, sketch the graph of  $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

## Sketch a quadratic from completed-square form

3 marks

## Turning point

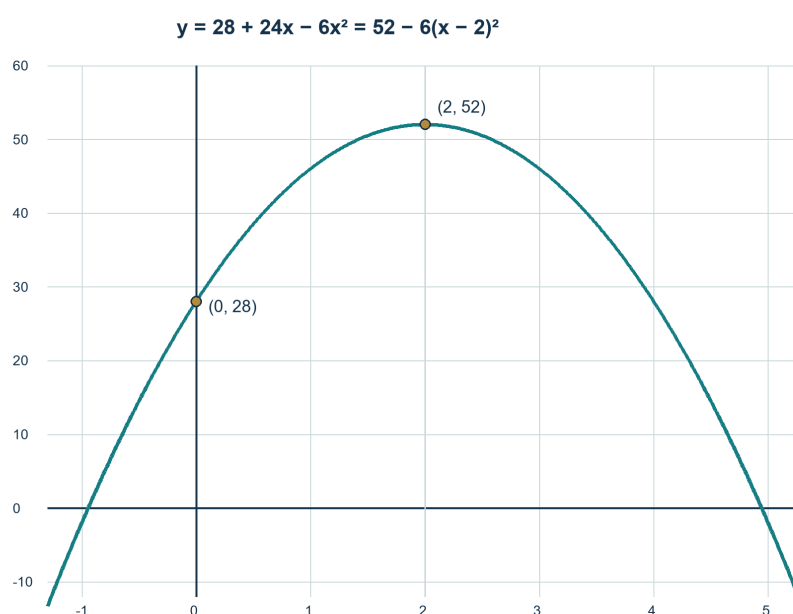
$$y = 52 - 6(x - 2)^2 \Rightarrow \text{maximum turning point } (2, 52).$$

## y-intercept

$$x = 0 \Rightarrow y = 28, \text{ so the graph crosses the } y\text{-axis at } (0, 28).$$

## Shape

The coefficient of  $(x - 2)^2$  is negative, so draw a symmetric downward parabola about  $x = 2$ .



Solved reference diagram

## Final answer

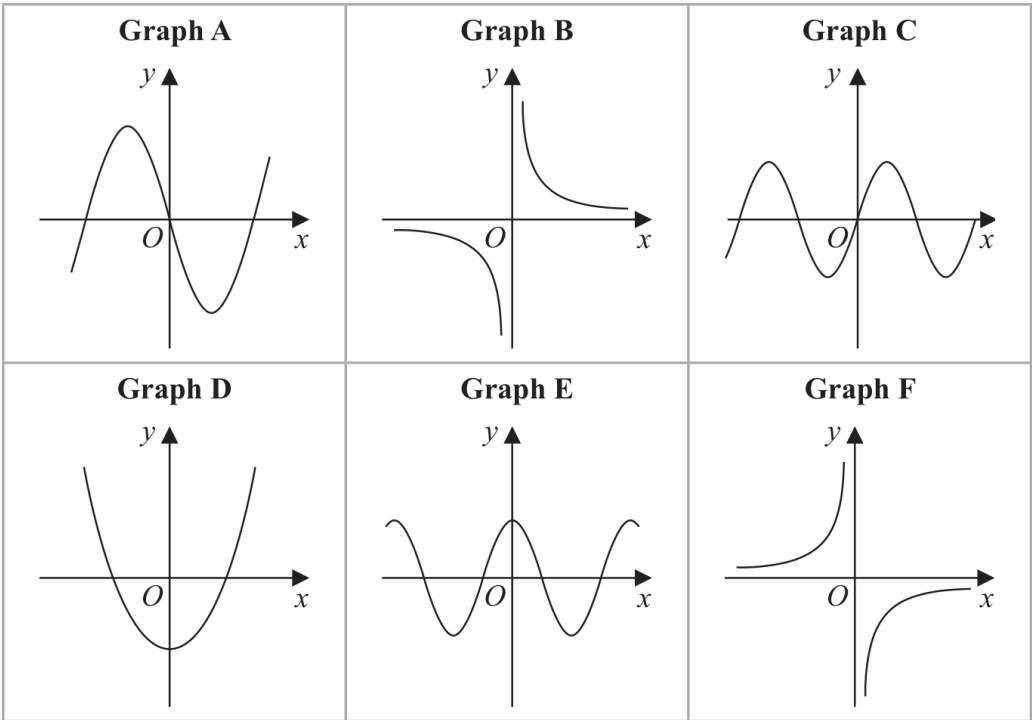
Downward parabola with turning point  $(2, 52)$  and  $y$ -intercept  $(0, 28)$ .

**Dependency check** Sequenced immediately after WU027; the printed equation is repeated, so the crop is self-contained.

4MA1-2H Q18

3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

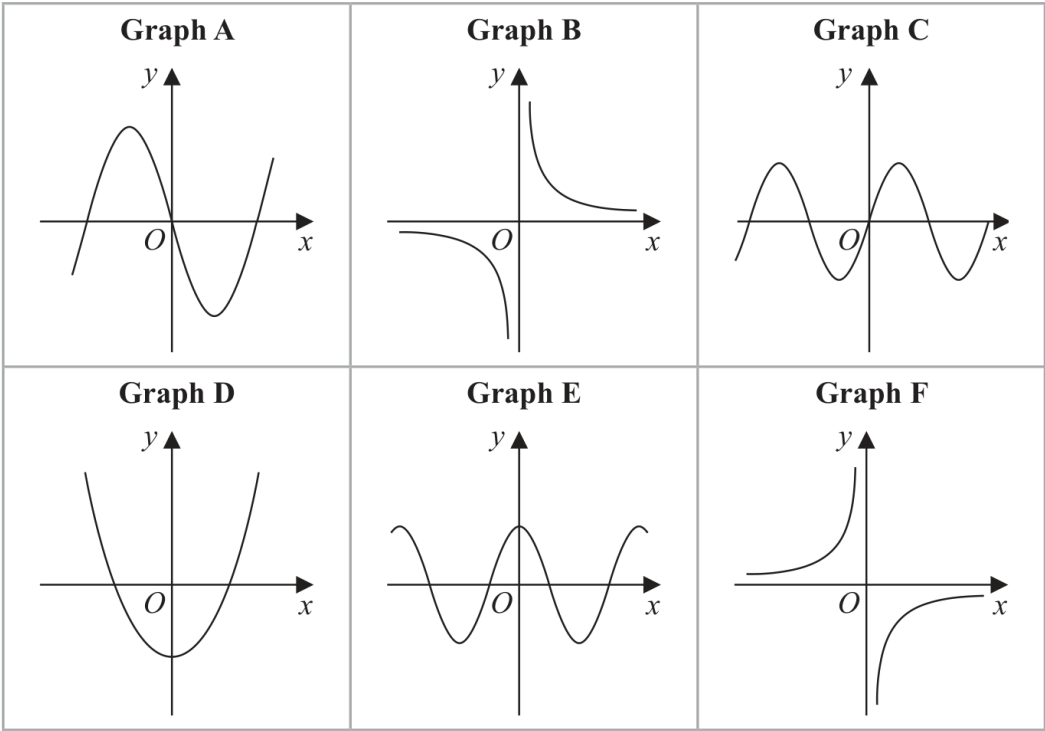
Write your answers on the dotted lines.

| Equation           | Graph |
|--------------------|-------|
| $y = \sin x$       | ..... |
| $y = -\frac{3}{x}$ | ..... |
| $y = 4x^3 - 5x$    | ..... |

**Continue your method****3 marks**

WORKING SPACE

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

| Equation           | Graph |
|--------------------|-------|
| $y = \sin x$       | ..... |
| $y = -\frac{3}{x}$ | ..... |
| $y = 4x^3 - 5x$    | ..... |

## Recognise function graph families

3 marks

## Match the sine graph

$y = \sin x$  passes through the origin and oscillates : Graph C.

## Match the reciprocal graph

$y = -3/x$  has positive branch for  $x < 0$  and negative branch for  $x > 0$  : Graph F.

## Match the cubic

$y = 4x^3 - 5x$  is an odd cubic with negative central gradient : Graph A.

## Final answer

$$\sin x \rightarrow C, \quad -3/x \rightarrow F, \quad 4x^3 - 5x \rightarrow A$$